“The Essentials of Computer Organization and Architecture”

Unit 1 – Exercises & Solutions

ESTiG-IPB (rufino@ipb.pt)

1.2.a. How many milliseconds are there in one second?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

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Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

Once 1 ms = 10^{-3} s ...

… then, in 1 s there are 1 / 10^{-3} = 10^3 = **1000 ms** = 1K ms

1.2.b. How many microseconds are there in one second?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
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Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
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Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.b. How many microseconds are there in one second?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
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Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
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Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

Once 1 us = 10^{-6} s ...

… then, in 1 s there are 1 / 10^{-6} = 10^6 = **1000000 us** = 1M us

1.2.c. How many nanoseconds are there in one millisecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
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Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

Once 1 ms = 10^{-3} s and 1 ns = 10^{-9} s

… then, in 1 ms there are $10^{-3} / 10^{-9} = 10^6$ = **1000000 ns** = 1M ns

1.2.d. How many microseconds are there in one millisecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

Once 1 ms = 10^{-3} s and 1 us = 10^{-6} s

… then, in 1 ms there are $10^{-3} / 10^{-6} = 10^3$ = **1000 us** = 1K us

1.2.e. How many nanoseconds are there in one microsecond?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

Once 1 us = 10^{-6} s and 1 ns = 10^{-9} s

… then, in 1 us there are $10^{-6} / 10^{-9} = 10^3$ = **1000 ns** = 1K ns

1.2.f. How many kilobytes are there in one gigabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

- **base 10**: once 1 KB = 10^3 B and 1 GB = 10^9 B
… then, in 1 GB there are $10^9 / 10^3 = 10^6$ = **1000000 KB** = 1M KB

- **base 2:** once 1 KB = 2^{10} B and 1 GB = 2^{30} B
… then, in 1 GB there are $2^{30} / 2^{10} = 2^{20}$ = **1048576 KB** = 1M KB

1.2.g. How many kilobytes are there in one megabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
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1.2.g. How many kilobytes are there in one megabyte?

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Answer:

- **base 10**: once 1 KB = 10^3 B and 1 MB = 10^6 B
… then, in 1 MB there are $10^6 / 10^3 = 10^3$ = **1000 KB** = 1K KB

- **base 2:** once 1 KB = 2^{10} B and 1 MB = 2^{20} B
… then, in 1 MB there are $2^{20} / 2^{10} = 2^{10}$ = **1024 KB** = 1K KB

1.2.h. How many megabytes are there in one gigabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
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1.2.h. How many megabytes are there in one gigabyte?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

- **base 10**: once 1 MB = 10^6 B and 1 GB = 10^9 B
… then, in 1 GB there are $10^9 / 10^6 = 10^3$ = **1000 MB** = 1K MB

- **base 2:** once 1 MB = 2^{20} B and 1 GB = 2^{30} B
… then, in 1 GB there are $2^{30} / 2^{20} = 2^{10}$ = **1024 MB** = 1K MB

1.2.i. How many bytes are there in 20 megabytes?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

1.2.i. How many bytes are there in 20 megabytes?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
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Yotta	Y	1 septillion = 10^{24}	2^{80}	Yocto	y	1 septillionth = 10^{-24}	2^{-80}

Answer:

- **base 10**: once 1 MB = 10^6 B
… then, in 20 MB there are 20 x 10^6 = **20000000 B**

- **base 2:** once 1 MB = 2^{20} B
… then, in 20 MB there are 20 x 2^{20} = **20971520 B**

1.2.j. How many kilobytes are there in 2 gigabytes?

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
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Answer:

- **base 10**: once 1 KB = 10^3 B and 1 GB = 10^9 B, then ...
  … in 2 GB there are 2 x 10^9 / 10^3 = 2 x 10^6 KB = **2000000 KB** = 2M KB

- **base 2:** once 1 KB = 2^{10} B and 1 GB = 2^{30} B, then ...
  ... in 2 GB there are 2 x 2^{30} / 2^{10} = 2 x 2^{20} KB = **2097152 KB** = 2M KB

1.2*. A RAM Memory of 256 GBytes of capacity has more ____ Bytes than a Hard Disk with 256 GBytes of capacity.

(present only the integer part of the number, without rounding, no use of scientific notation, no numerical powers (2^..., 10^...), no SI symbols (K, M, G, ...); for example, if the value obtained from the calculations is 123456,789, present only 123456).

Prefix	Symbol	Power of 10	Power of 2	Prefix	Symbol	Power of 10	Power of 2
Kilo	K	1 thousand = 10^3	$2^{10} = 1024$	Milli	m	1 thousandth = 10^{-3}	2^{-10}
Mega	M	1 million = 10^6	2^{20}	Micro	μ	1 millionth = 10^{-6}	2^{-20}
Giga	G	1 billion = 10^9	2^{30}	Nano	n	1 billionth = 10^{-9}	2^{-30}
Tera	T	1 trillion = 10^{12}	2^{40}	Pico	p	1 trillionth = 10^{-12}	2^{-40}
Peta	P	1 quadrillion = 10^{15}	2^{50}	Femto	f	1 quadrillionth = 10^{-15}	2^{-50}
Exa	E	1 quintillion = 10^{18}	2^{60}	Atto	a	1 quintillionth = 10^{-18}	2^{-60}
Zetta	Z	1 sextillion = 10^{21}	2^{70}	Zepto	z	1 sextillionth = 10^{-21}	2^{-70}
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Answer:

To measure RAM capacity, base 2 is used.
To measure Hard Drives capacity, base 10 is used.

A = RAM capacity = 256 x 2^30 bytes
B = Hard Drive capacity = 256 x 10^9 bytes

A – B = 256 x (2^30 – 10^9) = **18877906944**

1.3. System A performs a task in a time of the order of nanoseconds, and system B performs the same task in a time of the order of milliseconds. What is the order of magnitude of the acceleration (speedup) of A relative to B for the same task?

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Note:

If A spends time t_A on a task and B spends time t_B on the same task, the speedup of A relative to B is given by $\boldsymbol{t_B / t_A}$.

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Answer:

- the acceleration of A relative to B is given by
  t_B ms / t_A ns = (t_B / t_A) . (ms / ns) = (t_B / t_A) . (10^{-3} s / 10^{-9} s) = $(t_B / t_A)\ 10^6$

- although the specific value of the ratio t_B / t_A is not known, it is certain that it will be a value in the **order of millions** (10^6)

1.7.a. If a transistor currently takes an area of 2 um^2, what area will a transistor of the same type need 2 years from now, according to the 18-month version of Moore's Law? (present the result up to 4 decimal places, truncated)

1.7.a. If a transistor currently takes an area of 2 um^2, what area will a transistor of the same type need 2 years from now, according to the 18-month version of Moore's Law? (present the result up to 4 decimal places, truncated)

Answer:

- the Moore's Law 18-months version defines the **halving**, each 18 months, of the **dimensions** (area, volume) of the transistors

- however, this evolution is not linear; therefore, the simple "Rule of Three " cannot be used; instead, the evolution is exponential (with exponent <0) !

- if $\mathbf{d_0}$ is the current dimension, then the dimension $\mathbf{d_m}$ in **m** months will be

$$\mathbf{d_m = d_0 / 2^\wedge(m/18) = d_0 * 2^\wedge(- m/18)}$$

- in this scenario, d_0=2 um^2 and m=24 months (2 years), and so

$$\mathbf{d_{24} = 2 / 2^\wedge(24/18) = 0{,}7937\ um^2}$$

1.7.a* If an integrated circuit has currently 1 million transistors, how many transistors should it be possible to accommodate in a circuit of the same size, 2 years from now, according to the 18-month version of Moore's Law? (show only the integer part of the result)

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Answer:

- the Moore’s Law 18-month version defines the **doubling**, each 18 months, of the **number of transistors** in an equivalent circuit

- however, this evolution is not linear; therefore, the simple “Rule of Three ” cannot be used; instead, the evolution is exponential (with exponent >0) !

- if $\mathbf{n_0}$ is the current number of transistors, in **m** months that number will be

$$\mathbf{n_m = n_0 * 2\text{^}(m/18)}$$

- in this scenario, $n_0=1x10^6$ and m=24 months (2 years), and so

$$\mathbf{n_{24} = 1x10^6 * 2\text{^}(24/18) = 2519842{,}099789746 = 2519842 \text{ transistors}}$$

1.7.b. What is the influence of the validity of Moore's Law on a programmer's activity?

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Answer:

If it holds true, Moore's Law implies that the Information Technology (IT) industry will benefit from regular and predictable increases in performance at the CPU level.

Therefore, increasingly sophisticated and computationally demanding applications can be developed.